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Chapter 11

Basic Switch Configuration

This chapter is designed to help you prepare for the hands-on type of questions you will see on the ICND1 exam as it relates to day-to-day configuration of the switch and its ports. This chapter tests you on topics such as configure port speeds, duplex settings, and disabling ports. You also see questions surrounding topics such as VLANs and port security.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Configuring port speed and duplex settings
- Enabling and disabling ports
- Understanding port security and how to configure and troubleshoot port security
- Understanding and configuring VLANs

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Know how to look at the output of related `show` commands (such as show interface) and identify problems when looking at the configuration. For example, you should be able to tell the duplex setting, speed, and whether the port has been disabled.
- Know how to configure the speed and duplex settings on a port. You can use the speed command to configure the speed of the port or the duplex command to configure the duplex setting on the port.
- Know there are multiple steps to configuring port security. You first must make sure that the port is in access mode, and then you can configure the port security settings, such as the maximum number of addresses, the MAC address, and the action to perform upon address violation. Be sure to know this very well!
- Know how to configure port security with the sticky option, which learns the MAC address of the system connected to the port, instead of making you type the MAC address.
- Know how to create VLANs and configure an interface to be part of a VLAN. Always know the show commands to go with verifying your configuration.

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Configuring Ports

390–397 Choose the best answer(s).

390. Your manager would like a port description configured to port number 5 on the switch labeling the port as being used by the file server. What set of commands would you use?

(A) Use the following commands:

Switch#`interface f0/5`

Switch#`description File Server Port`

(B) Use the following commands:

Switch#`config term`

Switch(config)#`interface f0/5`

Switch(config-if)#`description File Server Port`

(C) Use the following commands:

Switch#`config term`

Switch(config)#`description File Server Port`

(D) Use the following commands:

Switch> `description File Server Port`

391. Jeff has disabled all of the unused ports on the switch. You would like to enable port 12. What command would you use?

(A) `disable shutdown`

(B) `enable port`

(C) `no shutdown`

(D) `enable no shutdown`

392. You wish to configure port 8 on the switch to negotiate its speed with the connecting system. What command would you use?

(A) `speed auto`

(B) `speed 100`

(C) `speed 10`

(D) `speed 1000`

393. You wish to configure port number 8 for full duplex. What command would you use?

(A) `duplex all`

(B) `duplex half`

(C) `duplex auto`

(D) `duplex full`

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394. You would like to configure all 24 ports on the switch for 100 Mbps. Which of the following represents the best way to do this?

(A) Use the following commands:

Switch(config)#`interface range f0/1 - 24`

Switch(config-if-range)#`speed 100`

(B) Navigate to each port and use the following command on each port:

`speed 100`

(C) Use the following commands:

Switch(config)#`interface f0/1 - 24`

Switch(config-if-range)#`speed 100`

(D) Use the following command:

Switch(config)#`speed 100 interface f01 - 24`

395. Which command sets the port speed to 100 Mbps?

(A) switch(config)#`speed 100`

(B) switch#`speed 100`

(C) switch>`speed 100`

(D) switch(config-if)#`speed 100`

396. What command would you use first to configure a setting on multiple ports on the switch?

(A) Use the `speed` command

(B) Use the `duplex` command

(C) Use `select ports` command

(D) Use the `interface range` command

397. You need to disable ports 6 to 12 on your switch. What commands would you use?

(A) Use the following commands:

Switch(config)#`interface range f0/6 - 12`

Switch(config-if-range)#`no shutdown`

(B) Use the following commands:

Switch#`interface range f0/6 - 12`

Switch#`shutdown`

(C) Use the following commands:

Switch#`shutdown interface range f0/6 – 12`

(D) Use the following commands:

Switch(config)#`interface range f0/6 - 12`

Switch(config-if-range)#`shutdown`

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Troubleshooting Switch Configuration

398–409 Choose the best answer(s).

398. What command shows you MAC addresses associated with each port on the switch?

(A) `show vlan`

(B) `show port-security addresses`

(C) `display mac-address-table`

(D) `show mac-address-table`

399. Looking at the figure below, what is the duplex setting of the port set to?

```
FastEthernet0/8 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0002.1604.3605 (bia 0002.1604.3605)
  Description: File Server Port
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00

(output has been omitted for briefness)
```

(A) Full duplex

(B) Half duplex

(C) Simplex

(D) No duplex

400. What command was used to create the output displayed in the figure below?

```
FastEthernet0/8 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0002.1604.3605 (bia 0002.1604.3605)
  Description: File Server Port
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00

(output has been omitted for briefness)
```

(A) `show interface f0/5`

(B) `show vlan`

(C) `show ip interface brief`

(D) `show interface f0/8`

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401. You wish to view the speed that port number 5 is using. What command would you use?

(A) `show speed`

(B) `show speed interface f0/5`

(C) `show interface f0/5`

(D) `show f0/5`

402. Looking at the figure below, what is the speed of the port set to?

```
FastEthernet0/8 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0002.1604.3605 (bia 0002.1604.3605)
  Description: File Server Port
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00

(output has been omitted for briefness)
```

(A) 10 Mbps

(B) 1000 Mbps

(C) 100 Mbps

(D) 10 Gbps

403. You are administering a 24-port switch that is divided into 4 VLANs. How many collision domains exist?

(A) 48

(B) 24

(C) 8

(D) 4

404. You are administering a 24-port switch that is divided into 4 VLANs. How many broadcast domains exist?

(A) 48

(B) 24

(C) 8

(D) 4

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405. Which of the following commands would you use to enable a port on the switch?

(A) `shutdown`

(B) `enable`

(C) `no shutdown`

(D) `no disable`

406. You are having trouble connecting to port 8 on the switch. You view the status of the port (shown in the figure below). What command would allow the port to function properly?

```
FastEthernet0/8 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0002.1604.3605 (bia 0002.1604.3605)
  Description: File Server Port
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00

(output has been omitted for briefness)
```

(A) `no shutdown`

(B) `shutdown`

(C) `ip address 12.0.0.10 255.0.0.0`

(D) `no cdp run`

407. Looking at the figure below, what is the bandwidth of the port set to?

```
FastEthernet0/8 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0002.1604.3605 (bia 0002.1604.3605)
  Description: File Server Port
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00

(output has been omitted for briefness)
```

(A) 100 Kbps

(B) 1000 Kbps

(C) 100000 Kbps

(D) 1000 Mbps

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408. When you connect the workstation to port 6, you can ping three other systems on the network, but you cannot seem to connect to the file server. You verify that others can connect to the file server, but you cannot connect to those other systems as well. What could be the problem?

(A) Wrong default gateway address.

(B) NAT is misconfigured.

(C) The port used by the workstation is disabled

(D) You are in a different VLAN than the file server.

409. You have a workstation connected to port 10 on the switch, but for some reason you cannot ping any other system on the network. You view the configuration of the port and get the output display in the figure below. What is likely the problem?

```
FastEthernet0/10 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0002.1604.3605 (bia 0002.1604.3605)
  Description: File Server Port
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00

(output has been omitted for briefness)
```

(A) You are using the wrong IP address.

(B) The port is disabled.

(C) The speed is 100 Mbps.

(D) ARPA is not being used.

Port Security

410–419 Choose the best answer(s).

410. What Cisco switch feature will allow you to control which systems can connect to a port on the switch?

(A) VTP

(B) Port Security

(C) VLANs

(D) STP

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411. What option allows you to configure a static MAC address on the switch by using the MAC of the connected system?

(A) static

(B) dynamic

(C) sticky

(D) usemac

412. Why would a network administrator configure Port Security on the switch?

(A) To filter packets by layer-3 addresses

(B) To translate private addresses to public addresses

(C) To prevent loops on the network

(D) To prevent unauthorized access to the network

413. Which of the following violation modes would block traffic not coming from the correct MAC address, but allow traffic from the specified MAC address?

(A) `shutdown`

(B) `restrict`

(C) `disable`

(D) `disconnect`

414. Which of the following actions disables the port when an address violation occurs?

(A) `disable`

(B) `shutdown`

(C) `disconnect`

(D) `restrict`

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415. What command was used to create the output displayed in the figure below?

```
Port Security              : Enabled
Port Status                : Secure-down
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 1
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
```

(A) `show interface f0/6`

(B) `show port-security interface f0/6`

(C) `show vlan`

(D) `show ip interface brief`

416. What mode must you place the interface into before you are able to configure Port Security on the interface?

(A) Access

(B) Trunk

(C) Dynamic

(D) Workstation

417. You are having trouble with a system connecting to port 5 on the switch. You want to see if the port has been configured for Port Security. What command do you use?

(A) `show mac-address-table`

(B) `show vlan`

(C) `show port-security interface f0/5`

(D) `show running-config`

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418. What command was used to create the output displayed in the figure below?

```
                        Secure Mac Address Table
-----------------------------------------------------------
Vlan    Mac Address       Type                    Ports
----    -----------       -----                   ------
1       1111.2222.3333    SecureConfigured        FastEthernet0/6
-----------------------------------------------------------


(output has been omitted for briefness)
Switch#
```

(A) `show mac-address-table`

(B) `show port-security address`

(C) `show ip interface brief`

(D) `show interface f0/8`

419. What are the three actions that you can configure when an address violation occurs? (Select three.)

(A) restrict

(B) disable

(C) shutdown

(D) disconnect

(E) protect

Configuring VLANs

420–427 Choose the best answer(s).

420. Your manager has heard a lot about the VLAN feature available on switches and is wondering what the benefit is. What would your response be?

(A) To create communication boundaries

(B) Filters traffic based off layer-3 address

(C) Filters traffic based off layer-4 address

(D) Used to prevent loops on the network

421. Which of the following switch features would you use to create multiple broadcast domains?

(A) STP

(B) VTP

(C) CDP

(D) VLANs

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422. You wish to view the VLAN configuration on the switch. What command do you use?

(A) `show port-security addresses`

(B) `show vlan`

(C) `show mac-address table`

(D) `show interfaces`

423. You are troubleshooting communication problems and suspect that a port is in the wrong VLAN. What command can you use to verify the VLANs that exist and what ports exist in each VLAN?

(A) `show vlan`

(B) `display all vlan`

(C) `show all vlan`

(D) `display vlans`

424. Which of the following protocols are used to carry VLAN traffic between switches? (Choose two.)

(A) VTP

(B) STP

(C) 802.1q

(D) ISL

(E) IGRP

425. What command would you use on a port to specify that it is allowed to carry all VLAN traffic across the port?

(A) `switchport mode trunk`

(B) `switchport mode vlan`

(C) `switchport mode access`

(D) `switchport mode vlanaccess`

426. Which of the following commands is used to create a VLAN name Floor1?

(A) `new vlan Floor1`

(B) `vlan Floor1`

(C) `create vlan Floor1`

(D) `vlan 2 name Floor1`

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427. Which of the following commands is used to place port 6 in VLAN 2?

(A) Use the following commands:

Switch(config)#`interface f0/6`

Switch(config-if)#`switchport access`

(B) Use the following commands:

Switch(config)# `switchport access vlan 2`

(C) Use the following commands:

Switch(config)#`interface f0/6`

Switch(config-if)#`switchport access vlan 2`

(D) Use the following commands:

Switch(config)#`interface f0/8`

Switch(config-if)#`switchport access vlan 2`